Course Code	**FOUNDATION ENGINEERING (PE – II)**	L	T	P	C
23A01604c		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. Understand the need for soil exploration and various methods used in site investigations.
2. Analyze the stability of slopes under different conditions using various stability methods.
3. Apply earth pressure theories to analyze retaining walls and soil pressures.
4. Evaluate the bearing capacity and settlement characteristics of shallow foundations.
5. Assess the load-carrying capacity and settlement of deep foundations, including pile and well foundations.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. Explain the principles of soil exploration, field testing, and soil investigation reporting.
2. Analyze slope stability using different failure theories and numerical methods.
3. Apply earth pressure theories to determine the stability of retaining walls.
4. Evaluate the bearing capacity and settlement of shallow foundations using theoretical and field methods.
5. Analyze deep foundations, including pile and well foundations, for their load-carrying capacity and settlement.

CO – PO Articulation Matrix

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	2	1	1	-	-	1	-	-	-	1	-	1	1	1
CO -2	2	2	1	1	2	-	-	-	-	-	-	1	1	1
CO -3	2	2	3	2	2	-	1	-	-	-	-	1	2	2
CO -4	2	1	2	2	3	1	-	-	-	-	-	1	2	1
CO -5	2	2	2	1	2	1	-	-	-	-	-	1	2	1

UNIT – I

SOIL EXPLORATION: Need - Methods of Soil Exploration - Boling and Sampling Methods — Field Tests Penetration Tests — Plate Load Test — Pressure Meter — Planning of Programme and Preparation of Soil Investigation Report.

UNIT – II

EARTH SLOPE STABILITY: Infinite and Finite Earth Slopes - Types Of Failures – Factor Of Safety of Infinite Slopes Stability Analysis by Swedish Arc Method, Standard Method Of Slices, Friction Circle Method — Taylor's Stability Number- Stability Of Slopes Of Earth Dams Under Different Conditions.

UNIT – III

EARTH PRESSURE THEORIES: Rankine's Theory of Pressure - Earth Pressures In Layered Soils — Coulomb's Earth Pressure Theory — Rebhann's and Cullman's Graphical Method.

RETAINING WALLS: Types Of Retaining Walls -Stability of Retaining Walls.

UNIT – IV

SHALLOW FOUNDATIONS: Types - Choice of Foundation Location of Depth Bearing Capacity — Terzaghi 'S, Meyerhoffs and Skempton's Methods
ALLOWABLE BEARING PRESSURE: Safe Bearing Pressure Based on N- Value Allowable Bearing Pressure; Safe Bearing Capacity and Settlement from Plate Load Test Allowable Settlements of Structures — Estimation of Consolidation Of Settlement.

UNIT – V

PILE FOUNDATIONS: Types Of Piles — Load Carrying Capacity of Piles Based on Static Pile Formulae — Dynamic Pile Formulae — Pile Load Tests Load Carrying Capacity Of Pile Groups in Sands and Clays — Settlement of Pile Groups.
WELL FOUNDATIONS: Types - Different Shapes of Wells - Components of Wells Functions And Forces on Well Foundation — Sinking of Wells Tilts and Shifts.

TEXT BOOKS:

1. Geotechnical Engineering by C.Venkataramaiah, New Age Pubilications(2002).
2. Soil Mechanics and Foundation Engineering by Arora, Standard Publishers and Distributors, Delhi 7th edition 2009
3. Soil Mechanics and Foundations by B.C.Punmia, Ashok Kumar Jain and Arun Kumar Jain, Laxmi publications Pvt. Ltd., New Delhi 17th edition 2017.

REFRENCE BOOKS:

1. Soil Mechanics and Foundation Engineering by Purushtoma Raj, Pearson Pubilications 2nd edition 2013
2. Principles of Foundation Engineering by Das, B.M., - (1999)–6th edition (Indian edition) Thomson Engineering
3. Foundation Engineering by Varghese,P.C., Prentice Hall of India., New Delhi.
4. Foundation Engineering by V.N.S.Murthy, CRC Press, New Delhi.

Online Learning Resources:

https://nptel.ac.in/courses/105105176